

Empirical Software Engineering

Write your answers directly on these pages; there's always a risk that loose papers disappear. Use the back also if possible.

On November 27 at 14.00–15.00 you are welcome to room 4123 in the EDIT house (Johanneberg), with questions about the grading.

Richard will be at the written exam after 15.30 to answer any questions.

“There is no end to education. It is not that you read a book, pass an examination, and finish with education. The whole of life, from the moment you are born to the moment you die, is a process of learning.”

— Jiddu Krishnamurti

Grade 3: 37 points; ~50%

Grade 4: 52 points; ~70%

Grade 5: 67 points; ~90%

Maximum: 74 points

Question 1 :

(8p) In *your* opinion, which **steps** are compulsory when conducting Bayesian data analysis with a focus on causal analysis? Please **explain** what steps one take when designing models, so that we ultimately can place *some* confidence in the results.

You can either draw a flowchart and explain each step, or write a numbered list explaining each step. (It's ok to write on the backside, if they haven't printed on the backside again...)

Solution: There is no standard. However, Gelman *et al.* (arXiv:2011.01808) have published a manuscript on arXiv explaining what they believed were the key parts in Bayesian data analysis. In short, a lot of the things they talk about in that paper are things we have covered in this course.

Start with a DAG and a null model, do prior checks, check diagnostics, do posterior checks, then conduct inferential statistics. Also, it would be good if you mention **comparisons** of models and that it is an **iterative** approach. Of course one could always add more...

Question 2 :

(9p) In Bayesian data analysis we have at least three principled ways of avoiding overfitting.

The **first** way makes sure that the model doesn't get too excited by the data. The **second** way is to estimate predictive accuracy. The **third**, and final way is to design models that actually tries do something about overfitting.

Which are the three ways? (3p)

Explain and provide examples for each of the three ways (2p+2p+2p)

Solution: In the first way we use regularizing priors to capture the regular features of the data. In particular it might be wise to contrast this with sample size! One could draw a figure to show this also, instead of explaining in text.

In the second way we rely on information theory (entropy and max_ent distributions) and the concept of information criteria, e.g., BIC, DIC, AIC, LOO, and WAIC. Explain and provide examples!

For the final, third way, the below is a good answer:

complete pooling

$$y_i \sim \text{Binomial}(n, p_i)$$

$$\text{logit}(p_i) = \alpha$$

$$\alpha \sim \text{Normal}(0, 2.5)$$

no pooling

$$y_i \sim \text{Binomial}(n, p_i)$$

$$\text{logit}(p_i) = \alpha_{\text{CLUSTER}[j]}$$

$$\alpha_j \sim \text{Normal}(0, 2)$$

partial pooling using hyper-parameters and hyper-priors

$$y_i \sim \text{Binomial}(n, p_i)$$

$$\text{logit}(p_i) = \alpha_{\text{CLUSTER}[j]}$$

$$\alpha_j \sim \text{Normal}(\bar{\alpha}, \sigma)$$

$$\bar{\alpha} \sim \text{Normal}(0, 1)$$

$$\sigma \sim \text{Exponential}(1)$$

Question 3 :

(4p) In the sciences we often differ between experiments and observational studies. What tension exists, concerning validity threats, between these two approaches (i.e., experiments and observational studies)?

Solution: The easiest way to answer this questions is to contrast internal validity in experiments with external validity in observational studies. Then one can do the same for external validity etc.

SOLUTION

Question 4 :

(8p) **Name** at least four distributions in the exponential family (4p). **Provide examples** of when one can use each distribution when designing statistical models and **explain** what is so special about each distribution you've picked, i.e., their assumptions (4p).

Solution: Take your pick, Gamma, Normal, Exponential, Poisson, Binomial, are some of the distributions... Each and every one of them come with their assumptions!

SOLUTION

Question 5 :

(8p) What is the **purpose** and **limitations** of using *Computer Simulations* and *Judgment Studies* as a research strategy? Provide **examples**, i.e., methods for each of the two categories, and **clarify** if one use mostly qualitative or quantitative approaches (or both).

Solution:**Judgement Studies:** Neutral setting

Judge or rate behaviors

Respond to stimulus

Discuss a topic of interest

Setting: Neutral setting

Actors: Systematic sampling

Goal: Seek generalizability over the responses not over the population of actors.

Responses are not specific to any context.

Unobtrusive and no control.

Examples of research methods:

Delphi study

Panel of experts

Focus group

Evaluation study

Interview studies

Qualitative and/or quantitative data

Computer Simulations:

Create a symbolic replica of a specific concrete system that can be executed by a computer.

Setting:

No empirical setting

Desk research.

Usually depends on prior empirical observations.

Low generalizability.

Goal: Evaluate many complex scenarios

Although the simulation responds to an Actor the response is not empirical. Result is as good as the model is accurate. Examples of research methods. Development of software that contains representation of the study variables. Usually, these variables are calibrated on empirical observations.

Example: Höst et al. To investigate bottlenecks and overload in RE processes. Setting: Nonempirical: a discrete event simulator was implemented in a DSL. Method: Four simulation scenarios with different parameter values to model different circumstances. Two ways were identified to avoid congestion: increase number of staff or productivity decrease the rate of new requirements e.g., through prioritization.

Question 6 :

(8p) Below follows an abstract from a research paper. Answer the questions,

- Which of the eight research strategies presented in the ABC framework does this paper likely fit? **Justify and argue!**
- What are the main validity threats of the paper, based on the research strategy you picked?
 - It would be very good if you can **list threats in the four common categories** we usually work with in software engineering.

Context: In software development organizations employing weak or collective ownership, different teams are allowed and expected to autonomously perform changes in various components. This creates diversity both in the knowledge of, and in the responsibility for, individual components.

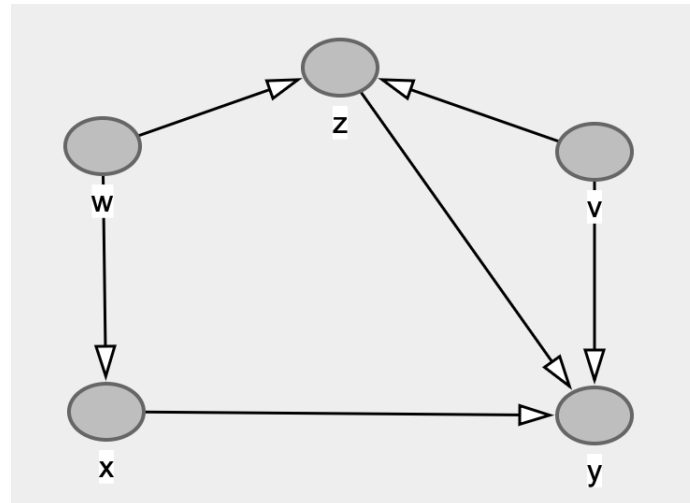
Objective: Our objective is to understand how and why different teams introduce technical debt in the form of code clones as they change different components.

Method: We collected data about change size and clone introductions made by ten teams in eight components which was part of a large industrial software system. We then designed a Multi-Level Generalized Linear Model (MLGLM), to illustrate the teams' differing behavior. Finally, we discussed the results with three development teams, plus line manager and the architect team, evaluating whether the model inferences aligned with what they expected. Responses were recorded and thematically coded.

...

Solution:

This is an observational study. It was not a study where we administrated the treatment. Hence a field study in practice. It has components of mining repositories but also qualitative elements.

**Question 7 :**

(7p) See the DAG above.

We want to estimate the **direct** causal effect of x on y . What if anything should we **condition on**? (2p)

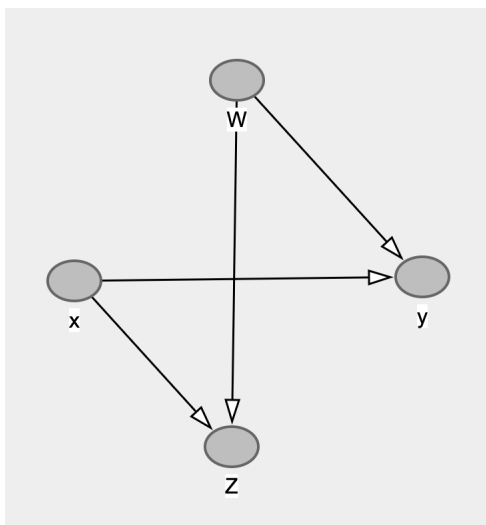
Design a **complete** model, in math notation, where the outcome y is a count $0, \dots, \infty$, and **include the variable(s)** needed to answer the above question. Also add, what you believe to be, **suitable priors on all parameters**. State any assumptions concerning your likelihood! (5p)

Solution:

We need to condition on $\{w\}$ or $\{v, z\}$ (if you can choose you'd pick the latter set, and you know why!)

Something like this would do then (since it's a positive count going towards infinity):

$$\begin{aligned}
 y_i &\sim \text{Poisson}(\lambda_i) \\
 \log(\lambda_i) &= \alpha + \beta_1 v_i + \beta_2 z_i + \beta_3 x_i \\
 \alpha &\sim \text{Normal}(0, 2.5) \\
 \beta_1, \beta_2, \beta_3 &\sim \text{Normal}(0, 0.2)
 \end{aligned}$$

**Question 8 :**

(7p) See the DAG above.

We want to estimate the **direct** causal effect of x on y . What if anything should we **condition on**? (2p)

Design a **complete** model, in math notation, where the outcome y is a count $0, \dots, \infty$ (however, the mean and the variance differs significantly!) **Include the variable(s)** needed to answer the above question. Also add, what you believe to be, **suitable priors on all parameters**. State any assumptions concerning your likelihood! (5p)

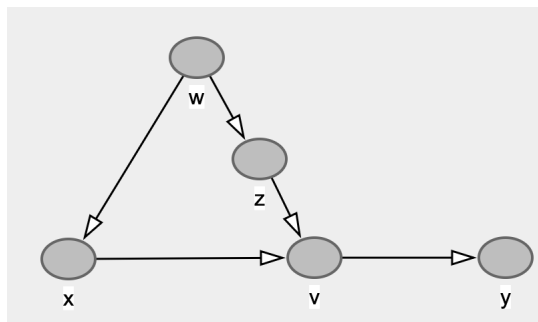
Solution:

We should not condition on anything! If we would foolishly condition on z or w then we need to also condition on w and z , respectively.

Something like this would do then:

$$\begin{aligned}
 y_i &\sim \text{Negative-Binomial}(\lambda_i, \phi) \\
 \log(\lambda_i) &= \alpha + \beta x_i \\
 \alpha &\sim \text{Normal}(0, 2.5) \\
 \beta &\sim \text{Normal}(0, 0.5) \\
 \phi &\sim \text{Exponential}(1)
 \end{aligned}$$

Instead of putting a prior on ϕ one could have another linear model with parameters and a logit link function to model the variability with more parameters.

**Question 9 :**

(7p) See the DAG above.

We want to estimate the **direct** and **total effect** of x on y . What if anything should we **condition on** for a) the direct effect and b) the total effect? (2p)

Design a **complete** model for estimating **the direct effect**, in math notation, where the outcome y is a real number (i.e., $\mathbb{R}$). **Include the variable(s)** needed to answer the above question. Also add, what you believe to be, **suitable priors on all parameters**. State any assumptions concerning your likelihood! (5p)

Solution:

This is a tricky one, since there's a difference between the direct and the total effect due to the pipe $x \rightarrow v \rightarrow y$.

For the direct effect, we should condition on $\{v\}$.

For the total effect, we should condition on $\{w, z\}$.

For the direct effect, given that we have a real number as an outcome, a Normal distribution is likely a good first choice,

$$y_i \sim \text{Normal}(\mu_i, \sigma)$$

$$\mu_i = \alpha + \beta_x x_i + \beta_v v_i$$

$$\alpha \sim \text{Normal}(0, 2.5)$$

$$\beta_x, \beta_v \sim \text{Normal}(0, 1)$$

$$\sigma \sim \text{Exponential}(1)$$

	PSIS	SE	dPSIS	dSE	pPSIS
m5.1	127.6	14.69	0.0	NA	4.7
m5.3	129.4	15.10	1.8	0.90	5.9
m5.2	140.6	11.21	13.1	10.82	3.8

Question 10 :

(8p) As a result of comparing three models, we get the above output. What does each column (PSIS, SE, dPSIS, dSE, and pPSIS) mean (5p)?

Having prediction in mind, which model would you select based on the output, and why (2p)?

Would your answer be different if I would have asked you to pick a model taking causality into account? (1p)

Solution:

One could argue that the first model, even though it leads by a slim margin, should be picked because it also has lower number of effective parameters. In reality, they only have a very little difference in PSIS (1.8) so no model is really significantly better/worse in out of sample prediction capability. What differs really is that m5.3 has more parameters and is thus penalized for it, but when we talk about few parameters, then it's hard to argue that it's that more complex, relatively speaking...

If we have a causal model and are only interested in causality, then we can throw the table away since the models might not represent our DAG.